RESEARCH PAPER | DATA ANALYTICS INTERNSHIP PROJECT

Customer Segmentation & Churn Pattern Analytics in European Banking

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Organisation

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Live Dashboard

https://europeanbankingcustomerchurnanalytics.streamlit.app/

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Dataset: 10,000 retail banking customers (France, Germany, Spain)

Abstract

Customer churn is one of the largest hidden costs in retail banking: every lost customer reduces lifetime value, raises acquisition cost and makes revenue less stable. This paper analyses a dataset of 10,000 retail customers of a European bank operating in France, Germany and Spain to answer three practical questions: **which customer groups churn most**, **how churn differs by geography, age and financial profile**, and **whether the bank is losing mainly low-value or high-value customers**. Customers were segmented by geography, age, credit score, tenure and balance, and churn rates, churn contribution and churned-versus-retained profiles were compared.

The overall churn rate is **20.37%** (2,037 of 10,000 customers). Churn is strongly concentrated: Germany churns at **32.4%** versus about 16% in France and Spain; customers aged 46-60 churn at **51.1%**; inactive members churn at **26.9%** versus 14.3% for active members; and customers holding three or four products almost always leave. Importantly, churned customers carry a **higher** average balance (EUR 91,109) than retained ones (EUR 72,745): customers with balances above EUR 100,000 account for **59.5% of all churners** and EUR **159.5 million** of balances walked out of the bank. The findings are delivered through an interactive Streamlit dashboard and translated into targeted retention recommendations.

Keywords: customer churn, customer segmentation, retail banking, exploratory data analysis, high-value customers, Streamlit.

1. Introduction

Banks routinely track a single churn percentage, yet a single number cannot tell a retention team where to act. Without segmentation, retention campaigns remain generic, reactive and expensive. Rich customer-level data allows a more precise approach: identify the segments with the highest churn, measure how much each segment contributes to total losses, and quantify the financial value at risk.

This project follows a segmentation-driven analytics approach. It measures overall and segment-level churn, compares European regions, studies engagement and tenure patterns, and isolates the high-value customers whose departure matters most. The results support strategic planning and marketing decisions.

1.1 Objectives

- Measure the overall churn rate and its distribution across customer segments.
- Compare churn behaviour across France, Germany and Spain.
- Understand churn among high-value customers and quantify revenue at risk.
- Evaluate engagement, product-holding and tenure patterns.
- Provide an interactive dashboard and actionable recommendations for decision-makers.

2. Dataset and Methodology

2.1 Dataset

The dataset contains 10,000 customer records with 13 usable fields: CreditScore, Geography (France, Germany, Spain), Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, the target variable **Exited** (1 = churned), plus identifier fields. Validation found **no missing values and no duplicate rows**; binary fields (HasCrCard, IsActiveMember, Exited) contained only 0 and 1, and Geography contained only the three expected countries.

2.2 Data preparation and segment design

The Surname field was removed as non-analytical. Four derived segmentation fields were created in Python (pandas) and used across the analysis and dashboard:

Dimension	Segments	Rule
Age band	<30, 30-45, 46-60, 60+	Age brackets
Credit score band	Low, Medium, High	<580, 580-740, >740
Tenure group	New, Mid-term, Long-term	0-2, 3-5, 6+ years
Balance segment	Zero, Low, High balance	0; up to 100,000; above 100,000 EUR

2.3 Analytical approach

- **Churn distribution:** overall rate, segment-wise rate and each segment's share of total churn.
- **Comparative analysis:** gender differences, geography x age interaction, churned versus retained profiles.
- **High-value analysis:** churn among customers with balance above EUR 100,000 and the balance leaving with them.
- **Tools:** Python, pandas, matplotlib/Plotly for analysis and visualisation; Streamlit for the dashboard.

3. Results and Exploratory Data Analysis

3.1 Overall churn

Of 10,000 customers, **2,037 exited**, giving an overall churn rate of **20.37%**: roughly one customer in five has left. The rest of the analysis explains where this churn is concentrated.

3.2 Churn by geography

Germany is the clear outlier. Although it holds only 25.1% of customers, it produces **39.96% of all churn** (814 customers), at a churn rate of 32.4%, double that of France (16.2%) and Spain (16.7%). France and Spain behave very similarly.

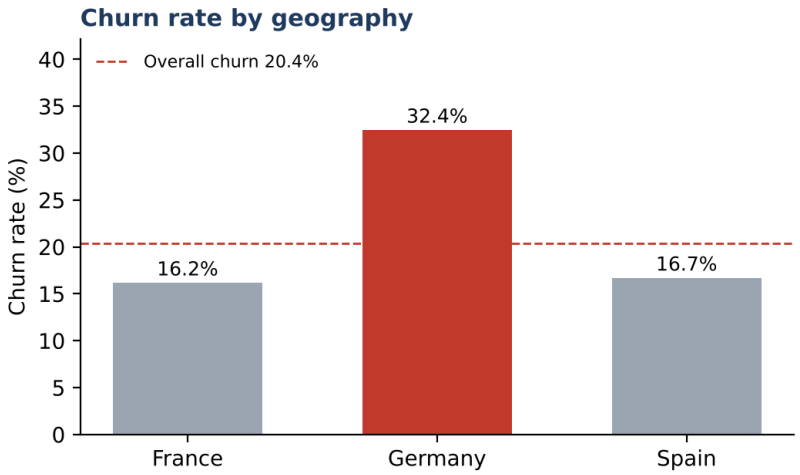


Figure 1. Churn rate by geography. Germany is roughly twice as high as France and Spain.

Geography	Customers	Churned	Churn rate	Share of all churn
France	5,014	810	16.15%	39.76%
Germany	2,509	814	32.44%	39.96%
Spain	2,477	413	16.67%	20.27%

3.3 Churn by age

Age is the strongest demographic driver. Customers aged **46-60 churn at 51.1%**, more than half, compared with 15.3% for 30-45 and only 7.6% for under-30s. The 46-60 group is just 16.5% of customers yet makes up 41.3% of churn. Churn falls again after 60 (24.8%).

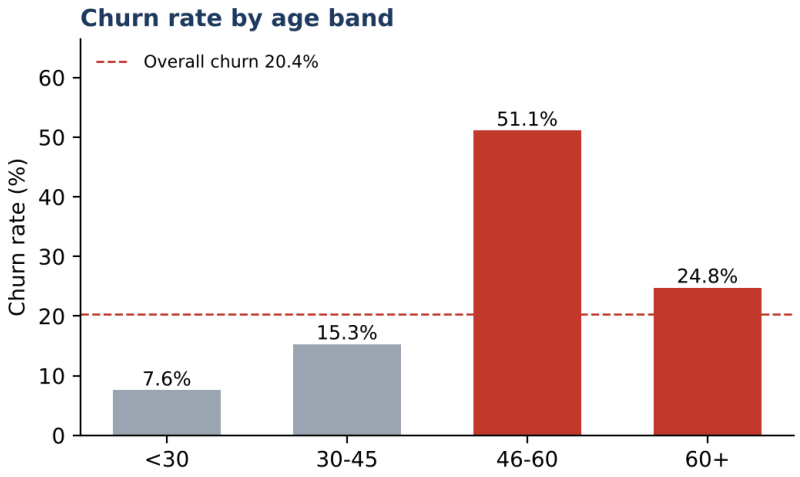


Figure 2. Churn rate by age band. The 46-60 group is the highest-risk age segment.

3.4 Geography and age together

The interaction shows where risk is most extreme: **German customers aged 46-60 churn at 67.3%**, about two out of three. Even German customers aged 30-45 churn at 25.3%, roughly double the same age group in France (11.9%) and Spain (12.5%). Customers under 30 are low-risk in every country.

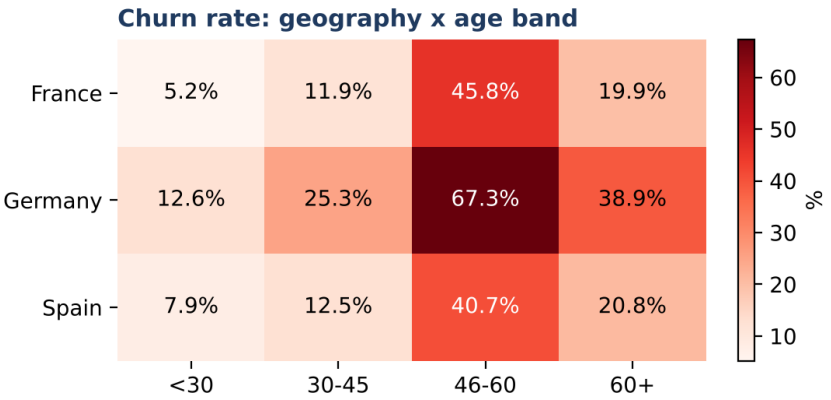


Figure 3. Geography x age band churn heatmap.

3.5 Gender

Female customers churn at **25.1%** versus **16.5%** for males, and women also make up 55.9% of all churners despite being 45.4% of customers. The gap holds in every country (for example 37.6% vs 27.8% in Germany), so it is a consistent pattern, not a one-country effect.

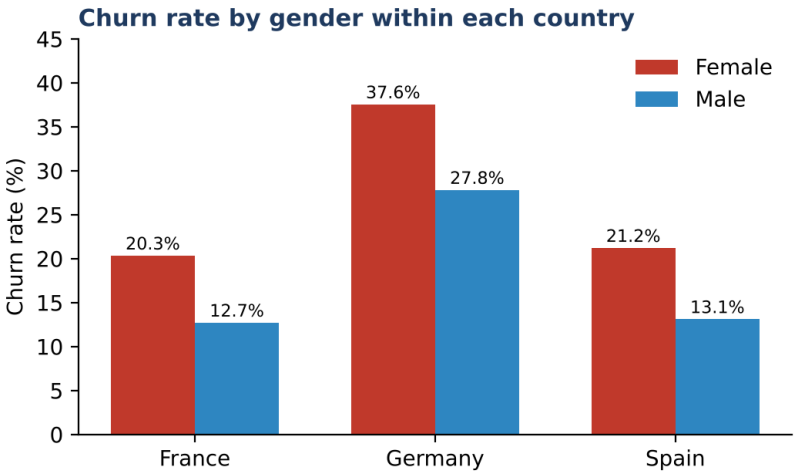


Figure 4. Female customers churn more than male customers in all three countries.

3.6 Engagement and product holding

Inactive members churn at **26.85%** versus **14.27%** for active members, a ratio of 1.88x, and account for 63.9% of churners. Product holding shows an even stronger, non-linear signal: customers with **two products** are the most loyal (7.6% churn), single-product customers churn at 27.7% (69.2% of all churners), while customers with **three products churn at 82.7% and with four products at 100%**. Having a credit card makes almost no difference (20.2% vs 20.8%).

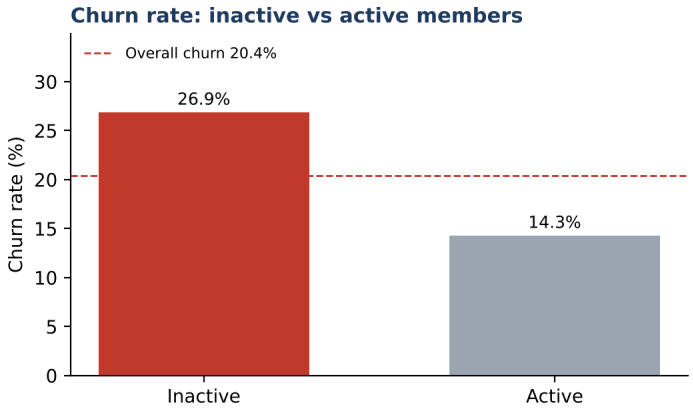


Figure 5. Inactive members churn almost twice as often as active members.

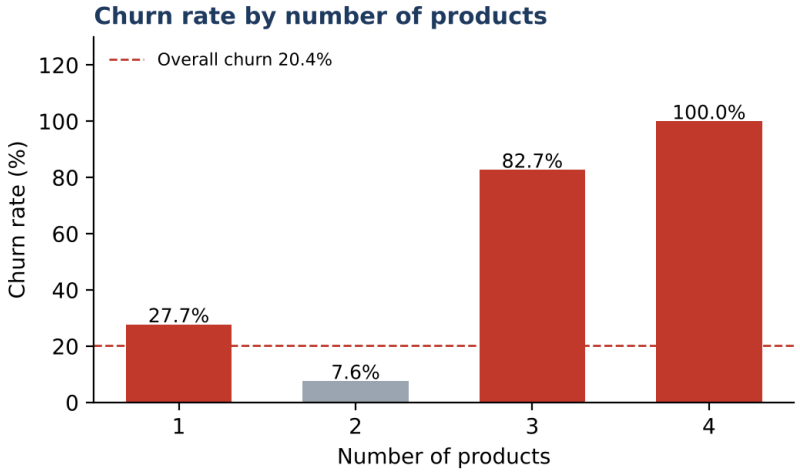


Figure 6. Churn by number of products. Two products is the loyalty sweet spot; three or more signals extreme risk.

3.7 Tenure and credit score

Tenure and credit score are weak differentiators. Churn is 21.2% for new, 20.8% for mid-term and 19.7% for long-term customers, so long tenure gives little protection. Credit-score bands range only from 19.8% (medium) to 22.0% (low). These dimensions are useful for sizing segments but are not good standalone targeting criteria.

Segment	Customers	Churn rate	Share of all churn
New (0-2 yrs)	2,496	21.15%	25.92%
Mid-term (3-5 yrs)	3,010	20.76%	30.68%
Long-term (6+ yrs)	4,494	19.67%	43.40%
Low credit score	2,362	22.02%	25.53%
Medium credit score	5,778	19.80%	56.16%
High credit score	1,860	20.05%	18.31%

3.8 Balance and high-value customers

Balance reveals the most financially important pattern. High-balance customers (above EUR 100,000) churn at **25.2%**, higher than low-balance (20.6%) and zero-balance customers (13.8%), and they contribute **59.5% of all churn** (1,211 customers). Salary is almost identical for churned and retained customers (EUR 101,466 vs 99,738); the difference is balance, which is far higher among those who leave.

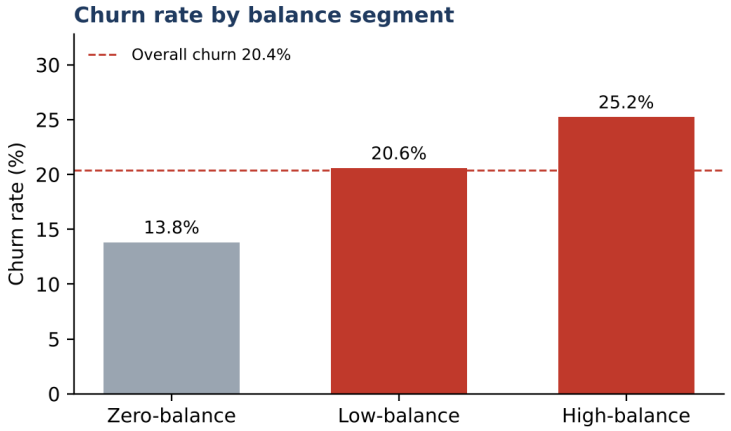


Figure 7. Churn rate by balance segment. High-balance customers are the most likely to leave among customers holding money.

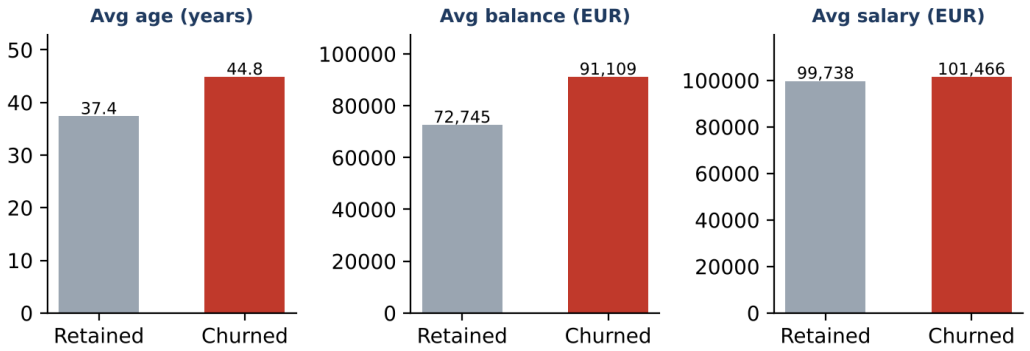


Figure 8. Churned versus retained customer profile. Churned customers are older and hold larger balances; salary is similar.

3.9 Revenue at risk

The 1,211 high-balance churners took **EUR 159,489,691** in balances with them. That is **85.9% of all balance lost to churn** (EUR 185.6 million) and 20.9% of the bank's total deposits in this sample (EUR 764.9 million). Churn here is not a volume problem; it is a value problem.

4. Key Performance Indicators

The following KPIs are computed in the dashboard and update dynamically when segment filters are applied:

KPI	Definition	Result
Overall Churn Rate	% of customers who exited	20.37%
Segment Churn Rate	Churn % within a segment	Germany 32.4%; Age 46-60 51.1%
High-Value Churn Ratio	Churn among customers with balance > 100,000 EUR	25.23% (1,211 customers)

KPI	Definition	Result
Geographic Risk Index	Share of churn / share of customers	Germany 1.59; Spain 0.82; France 0.79
Engagement Drop Indicator	Inactive churn / active churn	1.88x (26.85% vs 14.27%)
Revenue at Risk	Balance of high-value churners	EUR 159.5 million

A Geographic Risk Index above 1.0 means a region produces more churn than its customer share would suggest; Germany is the only region above 1.

5. Interactive Streamlit Dashboard

The analysis was operationalised as a live web application so non-technical stakeholders can explore the data themselves. The dashboard is available at https://europeanbankingcustomerchurnanalytics.streamlit.app/. Its modules are:

- **Overall churn summary:** KPI cards for churn rate, total customers, high-value churners, revenue at risk and inactive-churn gap.
- **Geography view:** churn by country and a geography x age heatmap.
- **Age and tenure comparison:** age-band, tenure-trend, tenure-group and credit-band charts.
- **High-value customer explorer:** sortable table of customers with balance above EUR 100,000.
- **Segment filters and dynamic KPIs:** sidebar filters for geography, gender, age band and membership status; all KPIs and charts refresh instantly for drill-down analysis.

6. Discussion and Recommendations

The data shows that churn is not random: it concentrates in identifiable, actionable segments. Because high-balance customers are leaving, the cost of churn is much larger than the 20% headline suggests. Recommended actions, ordered by expected impact:

- **1. Launch a German retention programme.** Germany combines the highest churn (32.4%) with the highest average balance (EUR 119,730 vs about EUR 62,000 in France and Spain). Investigate local competition, pricing and service quality, and prioritise relationship-manager outreach.
- **2. Target the 46-60 age group.** A 51% churn rate (67% in Germany) warrants dedicated offers such as wealth-planning, pension and mortgage-review services.
- **3. Protect high-balance customers first.** Build an early-warning list of customers above EUR 100,000 with declining activity and assign them proactive contact, since they represent 86% of balance lost.
- **4. Re-engage inactive members.** Inactivity doubles churn risk and is an observable early signal; trigger automated re-engagement (offers, app prompts, calls) when activity lapses.
- **5. Review the 3-4 product bundle.** Near-total churn for customers with three or four products suggests a product, pricing or service problem in cross-sold bundles; fix before pushing further cross-selling. Move single-product customers to a second product, where churn is lowest (7.6%).
- **6. Investigate the female-customer gap.** A consistent 8.6 percentage-point gap across all countries suggests an experience or product-fit issue worth qualitative research.

7. Limitations and Future Work

The dataset is a single-year snapshot (2025) and shows association rather than causation; the reasons customers leave (price, service, competitor offers) are not recorded. Segment thresholds such as the EUR 100,000 high-value cut-off and the credit-score bands are analyst-defined and could be tuned. Future work includes building a predictive churn model (logistic regression, random forest or gradient boosting) to score individual customers, adding time-series data to track churn trends, and estimating customer lifetime value to prioritise retention spend more precisely.

8. Conclusion

Segmentation-driven analysis turns a single churn figure into a clear retention agenda. About one in five customers leaves, but the risk is concentrated among German customers, those aged 46-60, inactive members, single-product and three-plus-product holders, and, most costly, customers with large balances. Targeting these segments is far more efficient than generic retention campaigns. The accompanying dashboard lets decision-makers monitor these KPIs and drill into any segment, supporting data-driven, targeted retention strategy.

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Live dashboard: https://europeanbankingcustomerchurnanalytics.streamlit.app/